

Aarish Khan

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TECHNICAL SKILLS

Languages:	Python, SQL
Data Analysis And Visualization	Pandas, NumPy, Matplotlib, Seaborn
Machine Learning:	Scikit-learn, TensorFlow, PyTorch
Financial Analytics:	Backtesting, Time Series Analysis, Risk Metrics

WORK EXPERIENCE

Jr.Quantitative Researcher

Mar 2023 - Present

Pace Stock Broking Services Pvt. Ltd., New Delhi, India

- Researched, backtested, and validated systematic trading strategies across equities, futures, and options using historical market data.
- Built and maintained historical OHLCV datasets, ensuring data quality and efficient retrieval for large-scale quantitative research.
- Analyzed strategy performance using Sharpe ratio, CAGR, maximum drawdown, win rate, profit factor, and risk-adjusted return metrics.
- Built an ML-based strategy clustering and selection engine to analyze 1,000+ systematic trading strategies and identify top-performing, low-correlated strategies based on return, risk, MTM behavior, and prevailing market conditions for improved portfolio diversification.
- Worked with quantitative developers to implement the backtested strategies.
- Generated research reports and performance analytics to support strategy selection and portfolio decisions.

Equity And Derivatives Trader

Nov 2022 - Mar 2023

Ebullient Securities Pvt. Ltd., Gurugram, India

- Executed trades manually on NEST with high speed and accuracy.
- Maintained daily trading diaries and end-of-day net position reports.

PROJECTS

1. ShoonyaOHLC AutoSync: Automated Nifty OHLC Data Pipeline [GitHub-Link](#)

Python — API Integration — Automation — GitHub Actions — Data Pipeline

- A code to auto-fetch daily OHLC data for Nifty Spot, Futures, and Options via Shoonya API.
- Automated end-to-end data pipeline using GitHub Actions cron jobs for daily post-market updates.
- Integrated Shoonya API + Google Drive API for reliable cloud-based historical data storage.
- Designed scalable folder structure and CSV datasets for efficient backtesting and analysis.
- Strengthened expertise in Python, API integration, automation, and cloud workflows.

2. Bank Customer Churn Prediction using ANN

[GitHub-Link](#)

Python — Machine Learning — Artificial Neural Networks — Streamlit — Model Deployment

- Built an Artificial Neural Network (ANN) model to predict whether a bank customer is likely to churn using customer demographic and financial features.
- Performed end-to-end data preprocessing including encoding categorical variables, feature scaling, and dataset splitting for robust training..
- Designed and trained a multi-layer ANN model, tuning hyperparameters to improve prediction performance.

- Deployed the trained model as an interactive web application using Streamlit, allowing users to input customer data and get real-time churn predictions..
- Demonstrated full ML lifecycle skills including data preparation, model building, evaluation, optimisation and production deployment.

EDUCATION

2026 - Present	Master of Computer Applications (MCA) from Sikkim Manipal University
2023 - 2025	Bachelor of Laws (LL.B) from Bundelkhand University, Jhansi
2020 - 2022	Bachelor of Arts in Economics from Sharda Devi Degree College, Jhansi
2017 - 2019	Mechanical Engineering Diploma from Govt. Polytechnic, Jhansi